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Unit 1 Mini Projects

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Project Report

Mini Project	Title
1.1	Simulating Link Budgets and Coverage Range with Python
1.2	Comparing FDD and TDD Resource Utilization Under Traffic Asymmetry
1.3	Validating and Extending the 0G–6G Generation Comparison (rule-based recommender option)

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Summary of key findings

- **Project 1.1.** With identical transmitter/receiver parameters, a 700 MHz link reaches **40×** farther than a 28 GHz link (1358.0 km vs 34.0 km under free-space loss), because 28 GHz suffers 20·log10(28000/700) = **32.04 dB** more path loss at any distance. The absolute ranges are unrealistically large because free-space loss ignores blockage, rain and other real losses.
- **Project 1.2.** At saturated load, FDD utilization falls from 100 % (50:50) to 60 % (90:10), while TDD with a 5 % guard stays at 95 %. TDD overtakes FDD once the downlink share exceeds **50 % + the guard fraction** (55 %, 60 %, 70 % for 5 %, 10 %, 20 % guard).

- **Project 1.3.** Fact-checking the generation rows (0G–6G, with 2.5G shown separately) against ITU, ETSI and 3GPP-linked sources found several peak-versus-typical gaps and date/definition nuances. A keyword recommender passed 12/12 (re-run after a vocabulary fix) and 7/8 on new sentences; the one miss and one false positive ("edge computing" → 2.5G) expose the limits of keyword matching.

0. Scope, environment and reproducibility

This report covers the three Unit 1 mini projects of the Mobile Communications lab manual. All results are produced by the three Python scripts submitted with the report (`project1_link_budget.py`, `project2_fdd_vs_tdd.py`, `project3_recommender.py`); every number quoted below comes from running those scripts. The scripts use Python 3 with NumPy and Matplotlib (project 1.3 uses only the standard library). Each script is run from a terminal as `python <script>.py` and writes its figures to the working folder.

Projects 1.1 and 1.2 are deterministic, so re-running them reproduces the numbers exactly. Project 1.3's fact-check relies on external sources consulted in October 2026; the 6G material in particular is still changing (Section 3.2).

1. Mini Project 1.1 – Link Budgets and Coverage Range

1.1 Objective

To implement the free-space path-loss (FSPL) formula and the link-budget equation from Section 1.10 in Python, and use them to compare received power and maximum usable range for a low band (700 MHz) and a millimetre-wave band (28 GHz) at equal transmit power.

1.2 Method

Two functions were implemented exactly as in the manual:

- `fspl_db(d_km, f_mhz) = 20·log10(d_km) + 20·log10(f_mhz) + 32.44`
- `link_budget_dbm(P_TX, G_TX, L_path, G_RX, L_other) = P_TX + G_TX − L_path + G_RX − L_other`

A distance array of 500 log-spaced points from 0.01 km to 100 km was used, as specified. Because the free-space range at 700 MHz exceeds that grid, the code also contains a closed-form solution (FSPL is invertible) and a supplementary 600-point grid to 10,000 km, used only for the "extended" figure. The same parameters were used for both bands so that any difference comes only from frequency.

Parameter	Value
Transmit power P_TX	40 dBm
Transmit antenna gain G_TX	15 dBi
Receive antenna gain G_RX	0 dBi
Other losses L_other	3 dB
Receiver sensitivity	−100 dBm
Reference distance	1 km
Bands	700 MHz (low-band) and 28,000 MHz (mmWave); 3.5 GHz added in Task 2

Table 1.1 – Fixed simulation parameters

1.3 Results

Band	Path loss @ 1 km	P_RX @ 1 km	Link margin @ 1 km	Max range, 100 km grid	Max range, closed form
700 MHz	89.34 dB	−37.34 dBm	62.66 dB	100.00 km (grid edge)	1358.0 km

Band	Path loss @ 1 km	P_RX @ 1 km	Link margin @ 1 km	Max range, 100 km grid	Max range, closed form
28 GHz	121.38 dB	−69.38 dBm	30.62 dB	33.66 km	34.0 km

Table 1.2 – Main comparison (output of project1_link_budget.py)

The range ratio is exactly **40.0×** (= 28000/700). The 700 MHz curve never falls below sensitivity on the manual's 100 km grid, so the grid-based maximum is simply the grid edge; this is why the closed-form range (1358.0 km) and the extended-range figure are reported as well.

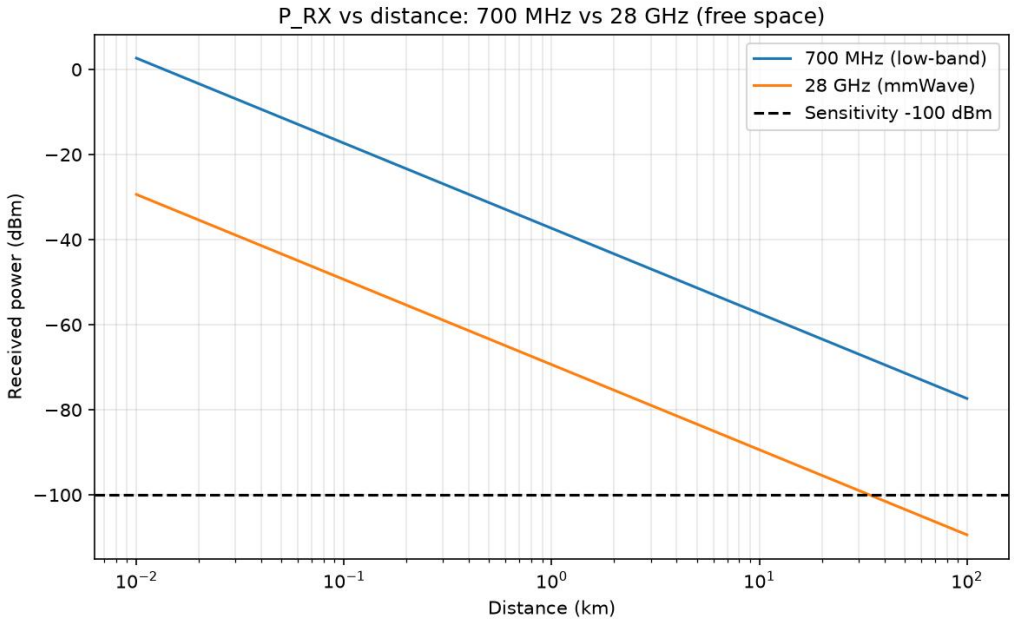


Figure 1.1 – P_RX versus distance (log axis) with the −100 dBm sensitivity line. Both curves are straight lines, as expected from a formula linear in log10(d).

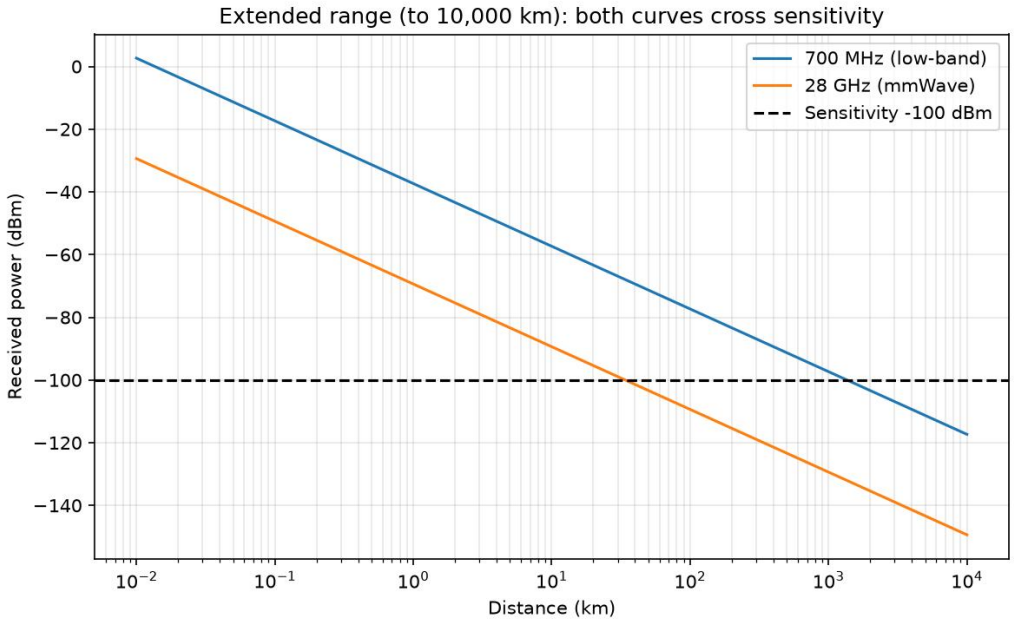


Figure 1.2 – Supplementary extended range (to 10,000 km): both curves cross the sensitivity line.

1.4 Analysis: why 700 MHz reaches farther

FSPL contains 20·log10(f). At equal distance the two bands therefore differ by 20·log10(28000/700) = 20·log10(40) = **32.04 dB** (121.38 − 89.34 dB, matching the table). Since every 20 dB of extra loss shrinks range by a factor of 10 on a log-distance axis, 32 dB of extra loss shrinks it by 10^(32.04/20) = 40×, about 1.6 orders of magnitude. This is the "one to two orders of magnitude" the manual predicts, and it is the physical reason low-band spectrum gives coverage while mmWave gives capacity but short range (Section 1.5).

Where the simulation departs from the manual's qualitative expectation. The manual expects mmWave to reach "at most a few hundred metres", whereas this simulation gives 34 km. The reason is that the model uses *only* free-space loss with a generous 55 dBm EIRP (40 dBm + 15 dBi) and −100 dBm sensitivity. Real mmWave links also lose power to blockage, foliage, rain and body shadowing, and typically use less favourable receiver sensitivity. Likewise, 1358 km at 700 MHz is far beyond a terrestrial radio horizon, so it should be read as a property of the idealised formula, not a deployable range. The *ratio* between bands is the robust result; the absolute ranges are upper bounds.

1.5 Lab tasks

Task 1 – 24 dBi array at 28 GHz

Raising G_TX from 15 to 24 dBi adds 9 dB, so range grows by 10^(9/20) = ×2.82: **34.0 km → 95.7 km**, recovering 61.7 km. The link is still about 14× shorter than 700 MHz (1358.0/95.7 = 14.2×), because the antenna gain recovers only 9 of the 32 dB frequency penalty.

Task 2 – adding 3.5 GHz

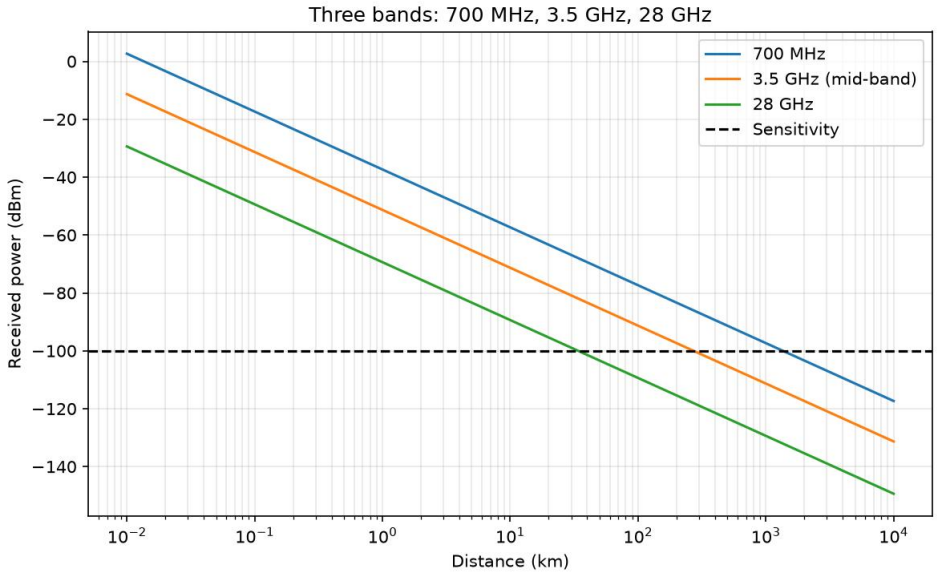


Figure 1.3 – 700 MHz, 3.5 GHz and 28 GHz on the extended grid.

The 3.5 GHz curve lies between the other two, with a maximum range of **271.6 km**: 5× shorter than 700 MHz (3500/700) and 8× longer than 28 GHz (28000/3500). This matches its role as a compromise between coverage and capacity.

Task 3 – 8 dB shadowing/fading margin

Band	Range before	Range after (L_other 3 → 11 dB)	Remaining	Range lost
700 MHz	1358.0 km	540.6 km	39.8 %	817.4 km

Band	Range before	Range after (L_other 3 → 11 dB)	Remaining	Range lost
3.5 GHz	271.6 km	108.1 km	39.8 %	163.5 km
28 GHz	34.0 km	13.5 km	39.8 %	20.4 km

Table 1.3 – Effect of an extra 8 dB margin

Every band keeps the same fraction, 10^(−8/20) = 39.8 %, because a fixed dB margin is a *multiplicative* range change independent of frequency. The absolute loss in kilometres is much larger for low-band, but the *relative* penalty is identical.

Task 4 – SNR and capacity (noise −104 dBm, 10 MHz)

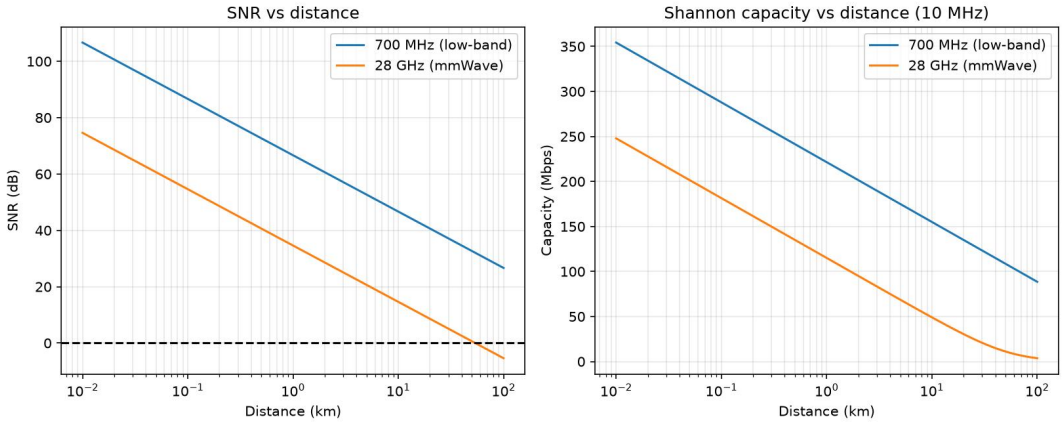


Figure 1.4 – SNR (left) and Shannon capacity C = B·log2(1+SNR) (right) versus distance.

Distance	700 MHz: P_RX / SNR / C	28 GHz: P_RX / SNR / C
0.1 km	−17.3 dBm / 86.7 dB / 287.9 Mbps	−49.4 dBm / 54.6 dB / 181.4 Mbps
1 km	−37.3 dBm / 66.7 dB / 221.4 Mbps	−69.4 dBm / 34.6 dB / 115.0 Mbps
10 km	−57.3 dBm / 46.7 dB / 155.0 Mbps	−89.4 dBm / 14.6 dB / 49.0 Mbps
30 km	−66.9 dBm / 37.1 dB / 123.3 Mbps	−98.9 dBm / 5.1 dB / 20.8 Mbps

Table 1.4 – SNR and capacity ceiling at selected distances

Between 0.1 km and 30 km, path loss rises by about 49.5 dB for both bands, yet 700 MHz capacity falls only 57 % (287.9 → 123.3 Mbps) and 28 GHz capacity falls 89 % (181.4 → 20.8 Mbps). Capacity degrades far more slowly than P_RX (which is linear in dB) because of the logarithm in Shannon's formula, and it degrades fastest once SNR becomes small (28 GHz beyond ~10 km), where the log2(1+SNR) term approaches a linear dependence on SNR. Caveat: these are Shannon *ceilings*; an SNR of 86.7 dB is idealised and real receivers saturate at much lower SNR, so the near-range values overstate achievable rates.

Task 5 – doubling transmit power versus doubling bandwidth (d = 20 km)

Band	SNR	Baseline	2× power (+3 dB)	2× bandwidth
700 MHz	40.6 dB	135.0 Mbps	145.0 Mbps (+7.4 %)	250.0 Mbps (+85.2 %)
28 GHz	8.6 dB	30.4 Mbps	39.5 Mbps (+29.9 %)	44.2 Mbps (+45.1 %)

Table 1.5 – Capacity gains (noise power is scaled by 3.01 dB when bandwidth is doubled)

Capacity is **linear in bandwidth** but only **logarithmic in power**. At high SNR, doubling power adds roughly one bit/s/Hz (log2 2 = 1) to a spectral efficiency of about 13.5 bit/s/Hz, i.e. +7.4 %. Doubling bandwidth doubles the number of Hz but also doubles the noise power (−3 dB SNR), and the net effect is still +85 %. At low SNR (28 GHz, 8.6 dB) the picture narrows: the system is power-limited, so

extra power helps more (+29.9 %) and extra bandwidth less (+45.1 %), since the lost 3 dB of SNR costs a larger share of a smaller spectral efficiency.

Task 6 – TDD guard periods and the link-budget duty cycle (qualitative)

A link budget gives the received power *while the transmitter is on*; it does not change with duty cycle. What changes is how much of the time the transmitter is on. An FDD base station transmits continuously on its downlink channel (duty cycle ≈ 100 % per direction). A TDD base station transmits only in its downlink slots and must also leave guard periods for switching and propagation delay, so its downlink duty cycle is (downlink slots/total slots) × (1 − guard fraction) and the same applies in the uplink. Consequences: (i) average transmitted power and heat are lower than the peak assumed in the budget; (ii) effective throughput is the instantaneous capacity multiplied by the duty cycle, so a link that "closes" on paper delivers less per direction on average; (iii) latency can increase because a packet may have to wait for its direction's next slot. These ideas are developed quantitatively in Project 1.2.

2. Mini Project 1.2 – FDD versus TDD under Traffic Asymmetry

2.1 Objective and background

To quantify how FDD and TDD utilize a fixed total capacity when traffic is asymmetric (downlink-heavy), connecting the qualitative trade-offs of Section 1.8 to numbers. FDD uses two **paired** channels with fixed, symmetric capacity; spare capacity in the under-used direction cannot be reallocated. TDD uses **one** channel divided into time slots, so the downlink/uplink ratio can follow demand, at the price of **guard-period overhead** and **network synchronization** (all neighbouring cells must agree on slot timing to avoid cross-interference).

2.2 Model

- **Normalised capacity:** the paired FDD spectrum, or the single TDD channel of equal total bandwidth, carries 100 capacity units.
- **FDD:** two always-on channels of 50 units each. Delivered traffic in each direction is min(demand, 50).
- **TDD:** a frame of 20 slots (5 units per slot). g slots per frame are guard/switching slots that carry no traffic; the remaining 20 − g slots are split between downlink and uplink as **whole slots**, chosen by the code to maximise delivered traffic for the given demand.
- **Demand:** a fraction of total capacity ("load"), split by the DL:UL ratio. Utilization = useful traffic delivered ÷ 100 units.
- **Scenarios:** 50:50, 70:30, 90:10 (required) plus 95:5; guard fractions 1/20 (5 %), 2/20 (10 %) and 4/20 (20 %).

2.3 Results (saturated network: offered load = 100 % of capacity)

DL:UL	FDD DL/UL (util.)	TDD 5 % guard	TDD 10 % guard	TDD 20 % guard
50:50	50.0 / 50.0 (100 %)	45.0 / 50.0 · 9D/10U (95 %)	40.0 / 50.0 · 8D/10U (90 %)	30.0 / 50.0 · 6D/10U (80 %)
70:30	50.0 / 30.0 (80 %)	65.0 / 30.0 · 13D/6U (95 %)	60.0 / 30.0 · 12D/6U (90 %)	50.0 / 30.0 · 10D/6U (80 %)
90:10	50.0 / 10.0 (60 %)	85.0 / 10.0 · 17D/2U (95 %)	80.0 / 10.0 · 16D/2U (90 %)	70.0 / 10.0 · 14D/2U (80 %)
95:5	50.0 / 5.0 (55 %)	90.0 / 5.0 · 18D/1U (95 %)	85.0 / 5.0 · 17D/1U (90 %)	75.0 / 5.0 · 15D/1U (80 %)

Table 2.1 – Effective throughput DL/UL (capacity units), TDD slot split (D = downlink, U = uplink) and utilization

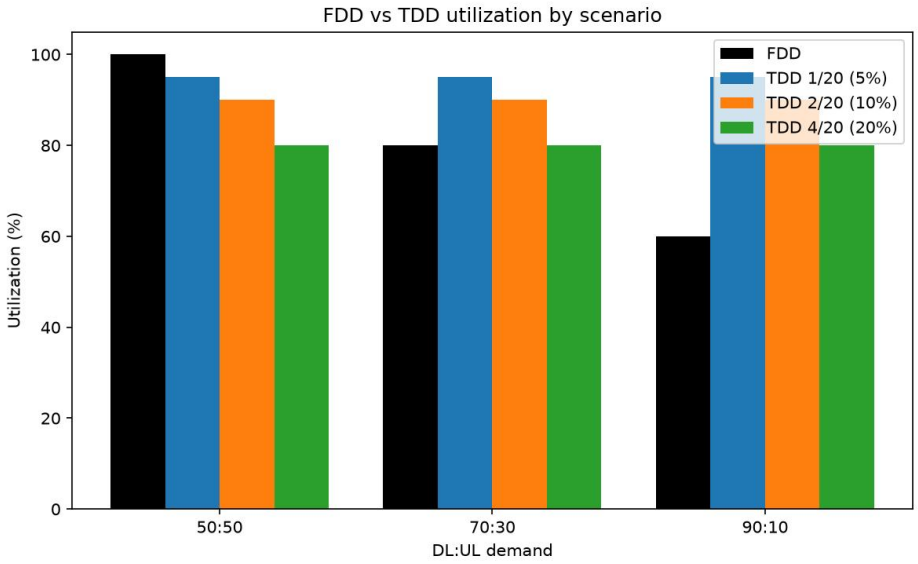


Figure 2.1 – Utilization by scenario for FDD and TDD with three guard fractions.

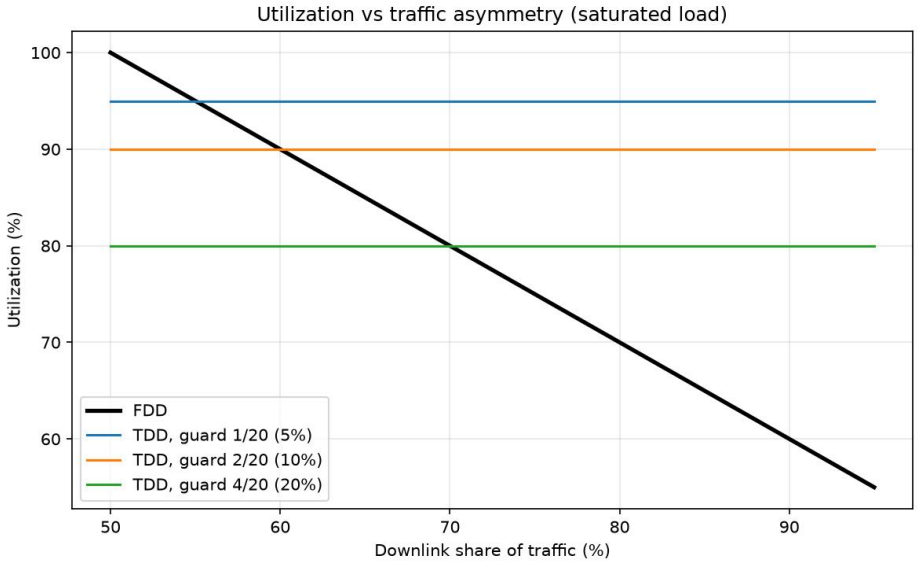


Figure 2.2 – Utilization versus downlink share (50–95 %) for FDD and three TDD guard settings.

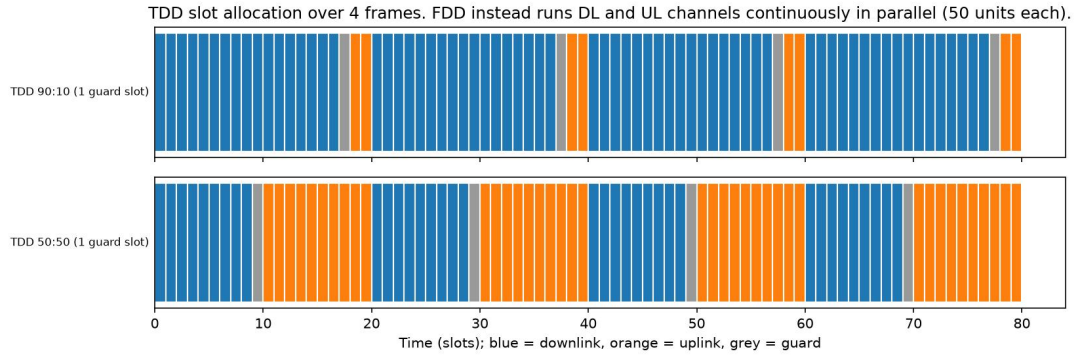


Figure 2.3 – (Optional extension) TDD slot allocation over four frames at 90:10 and 50:50 with one guard slot. FDD would instead run the downlink and uplink channels continuously in parallel.

2.4 Analysis

FDD is capped at 50 units per direction. Its utilization is 1.5 − s for a downlink share s > 0.5 (100 % at 50:50, 80 % at 70:30, 60 % at 90:10, 55 % at 95:5): the more asymmetric the traffic, the more uplink capacity sits idle while the downlink saturates. **TDD** re-balances slots, so its utilization is limited only by the guard overhead: 95 %, 90 % and 80 % for 5 %, 10 % and 20 % guard, independent of asymmetry in this saturated case.

Crossover. TDD beats FDD when 1 − g > 1.5 − s, i.e. when the downlink share exceeds **50 % + g**. The script reports the first sweep point above this (55.5 %, 60.5 %, 70.5 %), which is the analytical crossover (55 %, 60 %, 70 %) plus the 0.5-point sweep step. The 20 % guard curve touches FDD exactly at 70:30 (both 80 %). So:

- **FDD remains competitive** when traffic is close to symmetric: up to roughly 55 % DL with a 5 % guard, 60 % with 10 %, and 70 % with 20 %. At exactly 50:50, FDD is better by the full guard fraction (100 % vs 95 %).
- **TDD's advantage is largest at the highest asymmetry**: +15 points at 70:30, +35 points at 90:10 and +40 points at 95:5 with a 5 % guard (95 % vs 55 %, i.e. 1.73× the delivered traffic). Larger guard fractions erode this gap one-for-one (each extra 5 % of guard costs 5 points of utilization) and push the crossover to more extreme asymmetry.

Load matters. At 50 % offered load FDD and TDD both reach 50 % utilization in every scenario (script output), because FDD's downlink is not yet saturated. FDD can serve all traffic as long as the downlink demand stays at or below 50 units, i.e. up to a load of 0.5/s (about 71 % load at 70:30, 56 % at 90:10, 53 % at 95:5; derived from the model). TDD's flexibility only shows when the network is busy and asymmetric, which is exactly the downlink-dominated mobile case described in Section 1.8.

Beyond utilization (Section 1.8 trade-offs). FDD needs *paired* spectrum, which may be unavailable or costly; TDD needs only one channel and can exploit unpaired bands. TDD requires tight network-wide synchronization and guard periods (modelled here only as lost capacity); FDD avoids both and offers continuous transmission in each direction, which helps latency. The model deliberately ignores these latency and interference effects.

2.5 Limitations

- Capacity is normalised and idealised: no channel quality, scheduling losses, retransmissions, or asymmetric link budgets between DL and UL.
- Guard overhead is a fixed fraction of slots; real guard periods depend on cell size and frame configuration.
- Integer slots limit how closely TDD can match demand (e.g. 90:10 yields 17D/2U, delivering 85 of 90 downlink units demanded).
- Only steady-state throughput is modelled; the multi-frame figure is illustrative and does not model latency.

3. Mini Project 1.3 – Validating the 0G–6G Generation Comparison

3.1 Objective and chosen deliverable

To verify the generation summary of Section 1.2 against additional sources, identify the problems that motivated each generational shift, and build **option (b)**: a Python rule-based recommender that maps a plain-language communication requirement to the generation that first addressed that class of problem. This section presents the annotated fact-check (Section 3.2), the motivating problems (Section 3.3), the tool and its test results (Sections 3.4–3.5), and a synthesis (Section 3.6). Numbers in square brackets refer to the source list at the end.

Scope note. The claims checked below are those encoded in the recommender's justification strings, which summarise the Section 1.2 table (access technology, service, peak rate, key dates). Source quality varies: ITU, ETSI and 3GPP-presented material are authoritative; trade press, course slides and Wikipedia are used mainly to cross-check dates and to find disagreements, and are marked as secondary.

3.2 Annotated fact-check

Gen.	Claim checked	Finding / nuance	Sources
0G	MTS 1946 and IMTS 1964; first vehicle radio-telephone, few shared channels, operator-assisted connection, no cells.	Verified. MTS began in St. Louis on 17 June 1946; IMTS (1964) added channels but, per one source, kept direct dial. "Operator-assisted" therefore fits MTS but over-generalises to IMTS.	[15], [16]
1G	Analog cellular; NTT 1979, NMT 1981, AMPS 1983.	Verified, with a regional caveat. NTT (Tokyo, 1979) is cited as the first commercial cellular network; NMT launched 1981 and was first with international roaming; the first US commercial AMPS call was 13 Oct 1983 (Chicago). US-centric tables often call AMPS "the first". "FM" modulation was not independently checked.	[14], [15], [16], [22]
2G	GSM, 1991; digital, encrypted, SMS, roaming.	Verified; date detail disagrees. The first commercial GSM call is dated 1 June 1991 in one report and 1 July 1991 in another; the year is consistent. Encryption protects the phone–base-station link, not necessarily the whole network.	[12], [13], [23]
2.5G	GPRS/EDGE packet data.	Peak rates disagree. GPRS: 171.2 kbps theoretical (ETSI, 171 theoretical but 115 quoted as achievable); one wiki-derived source gives 40 kbps; EDGE: 384 kbps. A course slide gives *typical* rates of only 30–70 kbps, "similar to dial-up".	[10], [11], [21], [23]
3G	UMTS/HSPA; IMT-2000; 144 kbps – 2 Mbps; HSPA higher.	Verified. IMT-2000 targets: 144 kbps (vehicular), 384 kbps (pedestrian), 2 Mbps (indoor/fixed). HSPA+ (5 MHz) peaks at 14.4 → 28.8 → 43.2 Mbps in successive releases. The 2 Mbps figure is a single-user *indoor* peak and dominated early marketing. ETSI notes GPRS+EDGE already fall within the IMT-2000 range, so the 2.5G/3G boundary is blurry.	[7], [8], [9], [10], [24]

Gen.	Claim checked	Finding / nuance	Sources
4G	LTE/LTE-Advanced; IMT-Advanced; all-IP; tens to hundreds of Mbps.	Needs precision. IMT-Advanced targets 100 Mbps (high mobility) and 1 Gbps (low mobility). LTE Release 8 peaks at 300 Mbps DL / 75 Mbps UL, so first-release LTE was *not* formally IMT-Advanced, though widely branded "4G"; LTE-Advanced (Release 10) targets 1 Gbps DL / 500 Mbps UL.	[5], [6], [7]
5G	NR; ~1 ms URLLC latency; 1 million devices/km²; 20 Gbps peak; 100 Mbps user-experienced.	Verified, but these are minimum evaluation requirements. ITU-R M.2410: 20 Gbps DL / 10 Gbps UL peak; user-plane latency 4 ms (eMBB) and 1 ms (URLLC); 1,000,000 devices/km²; 100 Mbps DL user-experienced (dense urban). The peak is 200× the user-experienced rate, and the requirements are explicitly not a description of deployed performance.	[1], [2], [3], [4]
6G	IMT-2030; six usage scenarios; requirements in draft.	Verified, still moving. Framework M.2160 approved Nov 2023; six usage scenarios; example peak rates 50/100/200 Gbit/s; draft technical performance requirements (20 TPRs) completed Feb 2026 with formal approval expected at the Study Group in December; candidate submissions window opens Feb 2027. Values are targets, and 6G literature often quotes 1 Tbit/s.	[17], [18], [19], [20]

Table 3.1 – Fact-check of generation claims (all eight rows were checked, each against at least two sources)

Peak versus typical (the main discrepancy pattern)

Across generations, headline numbers are *peak* or *requirement* values, not what users get. GPRS (171.2 kbps theoretical) versus 30–70 kbps typical; 5G's 20 Gbps peak versus 100 Mbps user-experienced (the ITU defines the latter as the rate available to 95 % of users [20]); and 6G's proposed 50–200 Gbit/s peak versus 300–500 Mbit/s user-experienced examples [20]. A unit table that lists a single "peak data rate" per generation therefore hides the real gap.

3.3 Problems that motivated each generational transition

Transition	Motivating problems (beyond the summary table)	Sources
0G → 1G	1) Capacity: a handful of shared channels with no frequency reuse; IMTS added channels but "not nearly enough". 2) Coverage model: a single high-power transmitter served a whole service area (general engineering background, not from a cited source), whereas cells with low-power transmitters allowed frequency reuse and higher capacity.	[14], [16]
1G → 2G	1) Fragmentation: incompatible national analog systems (AMPS, NMT, TACS, NTT) limited roaming; NMT was the early exception. 2) Security: analog calls could be overheard; 2G added digital encryption (radio link). 3) Spectrum efficiency: 2G fit more users per band.	[14], [23]
2G → 2.5G	1) Circuit-switched data is wasteful: CSD gave 9.6 kbps and tied up a time slot for the whole session; HSCSD (up to 64 kbps) occupied several slots, was priced higher and "never enormously popular". 2) Bursty traffic (web, email) needs shared, volume-billed packet transport, so GPRS added the SGSN and GGSN nodes.	[10], [11]
2.5G → 3G	1) Rates: GPRS/EDGE peaks of 171/384 kbps theoretical and much lower typical rates versus IMT-2000's 144 kbps/384 kbps/2 Mbps targets designed around data. 2) Divergent 2G families: ITU IMT-2000 defined a global framework with five radio interfaces.	[8], [9], [10], [21]
3G → 4G	1) Legacy core: 4G removed circuit switching in favour of an all-IP network. 2) Capacity for rich media: HSPA/HSPA+ peaks of 14.4–43.2 Mbps versus IMT-Advanced's 100 Mbps/1 Gbps targets.	[5], [22], [24]

Transition	Motivating problems (beyond the summary table)	Sources
4G → 5G	1) Latency: ITU targets of 4 ms (eMBB) and 1 ms (URLLC) for control-type services. 2) Scale and diversity: 1,000,000 devices/km² (mMTC) and 20 Gbps peak (eMBB) with one framework, motivating the three service categories.	[1], [2]
5G → 6G	1) New capabilities: integrated sensing and AI-and-communication, absent from earlier IMT scenarios. 2) Higher targets: peak 50–200 Gbit/s examples, user-experienced 300/500 Mbit/s and mobility of 500–1000 km/h, plus an immersive-communication scenario.	[17], [19], [20]

Table 3.2 – At least two motivating problems per transition

A caution on "first generation to solve it"

The mMTC problem illustrates why the recommender's mapping is only approximate. The ITU category of massive connectivity (1 million devices/km²) is a 5G requirement, yet an ITU article cited in the IMT-2020 reference list is titled "LTE-M and NB-IoT meet the 5G performance requirements" [25]. I did not open that article, so the claim is treated as a lead, not a fact; but if confirmed, cellular IoT was practically addressed by 4G-era technologies before 5G New Radio.

3.4 The recommender tool

`project3_recommender.py` implements the dictionary-based approach suggested by the manual (no machine learning, standard library only).

- **Rules:** eight entries in chronological order (0G, 1G, 2G, 2.5G, 3G, 4G, 5G, 6G), each with a problem class, a keyword list and a justification string. Keywords ending in * are stem matches (e.g. `robot*`); all others are whole words/phrases.
- **Normalisation:** text is lower-cased and hyphens/slashes/underscores become spaces, so "low-latency" matches "low latency".
- **Scoring:** the generation with the **most distinct keyword hits** wins. **Ties** are reported as ambiguous and the *earliest* generation is returned, because the question asks which generation *first* addressed the problem. Other partial matches are listed as notes.
- **No match:** if no keyword fires, the tool returns "No clear match", stating that the need may be generation-independent (coverage, battery life, price).
- **Use:** `python project3_recommender.py` runs the built-in tests; `python project3_recommender.py -i` starts an interactive prompt.

3.5 Test results

The tests were written with expected answers before running. Round 1 contains 12 sentences, including a deliberate no-match case and a case that spans two generations. According to the script's own labelling, Round 1 was **re-run after the keyword list was fixed**, so its 12/12 reflects a tuned vocabulary and is not independent evidence. Round 2 contains eight *new* sentences and is the fairer estimate.

#	Round 1 input	Expected	Result	Matched keywords
1	Make a phone call while walking down the street	1G	1G ✔	phone call
2	Stop strangers listening in on my calls and let me send a quick text to a friend	2G	2G ✔	listening in
3	Check my email on my phone and pay only for the data I use, not the minutes	2.5G	2.5G ✔	email on my phone
4	Video chat with my grandmother and browse the web on a handheld device	3G	3G ✔	video chat, browse*

#	Round 1 input	Expected	Result	Matched keywords
5	Stream a high-definition movie on a train	4G	4G ✔	high definition, stream*
6	Remotely operate a surgical robot with almost no delay	5G	5G ✔	robot*
7	Let a hundred thousand parking sensors report once an hour across a city	5G	5G ✔	sensor*
8	Walk into a virtual meeting room and interact with a hologram of a colleague	6G	6G ✔	hologram*
9	Reach someone using a vehicle radio and an operator who connects the call by hand	0G	0G ✔	vehicle radio
10	Improve coverage and battery life in a remote village	No clear match	No clear match ✔	—
11	Use the same number abroad and keep my calls private	2G	2G ✔	abroad, calls private
12	Send a photo message and watch a short video clip on my phone	2.5G or 3G	2.5G ✔	photo message

Table 3.3 – Round 1: 12/12 as expected (after vocabulary fix)

#	Round 2 input (new)	Expected	Result	Matched keywords
1	Travel overseas and keep using my phone without changing the SIM	2G	2G ✔	sim, overseas
2	Browse the news on my phone while commuting	3G	3G ✔	browse*
3	Control a fleet of warehouse robots with split-second reactions	5G	5G ✔	robot*
4	Share a mobile hotspot with my laptop for a video conference	4G	4G ✔	hotspot
5	Send a picture message to my brother over a packet data network	2.5G	2.5G ✔	packet*, picture message
6	Detect moving objects using the same radio signals that carry my data	6G	No clear match ✘	—
7	Lower my monthly phone bill	No clear match	No clear match ✔	—
8	Play online games with friends and read data from temperature sensors	4G or 5G	4G ✔ (tie 4G/5G flagged)	online gam* (4G); sensor* (5G)

Table 3.4 – Round 2: 7/8 as expected

Failure analysis and additional probing

- **Miss (Round 2, #6).** The sentence describes integrated sensing in everyday words; the 6G rule only knows the technical phrases “integrated sensing” and “sensing and communication”. This is the central weakness of keyword matching: it cannot recognise paraphrases. The same idea phrased technically (“Detect moving objects using integrated sensing and communication”) was correctly mapped to 6G in an extra probe.
- **False positive (extra probe).** “Run edge computing workloads for my factory” returned **2.5G**, because the keyword edge (EDGE data technology) matches the unrelated word “edge”. Keyword lists need context-aware phrases (e.g. “edge data rates”) to avoid this.
- **Correct on a spanning case.** The two-need sentence (online games + temperature sensors) was flagged as a 4G/5G tie and the earliest generation returned, as designed. The tool returns one answer, so a user must read the note to see that two generations apply.
- **Other probes.** “Control a robotic arm with low latency” → 5G; “Stream music while jogging” → 4G (though music streaming was feasible earlier, a limitation of “first generation” semantics); “Send a text message” → 2G.

With only 20 scripted sentences, these percentages are descriptive, not a reliable accuracy estimate.

3.6 Synthesis: where real data nuanced the simplified table

The clearest nuance is the peak-versus-typical gap. The recommender's 5G justification quotes "20 Gbps peak / 100 Mbps user-experienced" side by side, which preserves the distinction the simple table loses. Real data complicated three other claims: (1) "first" is regional: NTT in 1979, NMT in 1981 and AMPS in 1983 are all legitimate "first 1G" answers; (2) "4G = LTE" hides that first-release LTE (300 Mbps) did not meet IMT-Advanced's 1 Gbps; and (3) the 2.5G/3G boundary is blurred because GPRS+EDGE reached the IMT-2000 range [10]. Finally, the 6G row is unusually unstable: as of the March 2026 ITU release the 6G performance requirements existed as a completed draft awaiting formal approval, so any value quoted for 6G is a target and should be re-checked.

As a communication tool, the recommender answers "which generation first addressed this?" quickly and transparently (it shows matched keywords and a justification), and its "No clear match" and tie notes are honest about uncertainty. Its limits are vocabulary coverage, false positives from ambiguous words, and the fact that "first generation to solve it" is a simplification of overlapping technologies.

4. Overall conclusions

- **Physics sets the coverage/capacity trade-off.** A fixed 32 dB frequency penalty makes 28 GHz reach 40× less far than 700 MHz; antenna gain and bandwidth/power choices can recover only part of it, and real impairments (blockage, fading margins) shrink every band by the same factor.
- **Duplexing is a utilization decision.** FDD wastes capacity as traffic becomes asymmetric; TDD pays a constant guard overhead but follows demand. The break-even is a downlink share of 50 % + the guard fraction, and TDD's benefit is largest for heavily downlink-dominated, heavily loaded networks.
- **Summary tables simplify.** Headline generation figures are peaks or requirements, dates are regional, and boundaries between generations blur. Tools built on such tables (like the recommender) inherit these simplifications and should expose their uncertainty.

Sources

Sources consulted October 2026. Items marked (secondary) are encyclopaedic, trade-press or course-slide material used for cross-checking.

[1] ITU-R WP 5D, "Requirements for IMT-2020" workshop slides (summarising Report ITU-R M.2410): 20 Gbps DL peak, 1,000,000 devices/km². https://www.itu.int/en/ITU-R/study-groups/rsg5/rwp5d/imt-2020/Documents/ (S01-1 Requirements for IMT-2020 Rev.pdf)

[2] TelecomTV, "ITU agrees on key 5G performance requirements for IMT-2020" (secondary). https://www.telecomtv.com/content/5g/itu-agrees-on-key-5g-performance-requirements-for-imt-2020-14401/

[3] Utility Dive (sponsored), "Where do cellular generational standards come from, and what is 6G?" – quotes ITU-R M.2410-0 targets (secondary). https://www.utilitydive.com/spons/where-do-cellular-generational-standards-come-from-and-what-is-6g/698651/

[4] ISPreview, "Arqiva unveil first UK trial of 5G mobile as ITU set draft performance spec," Feb 2017 – 100 Mbps DL user-experienced (secondary). https://www.ispreview.co.uk/index.php/2017/02/arqiva-unveil-first-uk-trial-5g-mobile-itu-set-draft-performance-spec.html

[5] T. Nakamura (3GPP TSG-RAN Chairman), "Proposal for Candidate Radio Interface Technologies for IMT-Advanced Based on LTE Release 10 and Beyond (LTE-Advanced)," 3GPP IMT-Advanced Evaluation Workshop, Beijing, Dec 2009 (slides). https://www.slideshare.net/slideshow/rev-0900023gpplte-advancedintroduction-100324021148-phpapp01/5330124

[6] Wikipedia, “4G” – IMT-Advanced requirements; LTE Release 8 and branding (secondary). https://en.wikipedia.org/wiki/4G

[7] M. Rumney (Agilent), “IMT-Advanced: 4G Wireless Takes Shape in an Olympic Year” – IMT-2000 peak data rates and peak-versus-average gap. https://moodle.upm.es/en-abierto/pluginfile.php/190/mod_label/intro/IMT-Advanced_Rumney.pdf

[8] “Wireless and Cellular Telecommunications,” 3rd ed., Ch. 6 “3G Systems” (GlobalSpec) – IMT-2000 144 kbps/384 kbps/2 Mbps. https://www.globalspec.com/reference/81103/203279/chapter-6-3g-systems

[9] Wikipedia, “IMT-2000” – five radio interfaces (ITU-R M.1457) (secondary). https://en.wikipedia.org/wiki/IMT-2000

[10] ETSI, “GSM / 2G technologies” – GPRS 115 kbps (171 theoretical), EDGE 384 kbps, HSCSD, IMT-2000 range. https://etsi.org/technologies/mobile/2g

[11] CommsBrief, “What is the difference between GPRS, EDGE and EGPRS?” (secondary). https://commsbrief.com/what-is-the-difference-between-gprs-and-edge/

[12] Network World, “First commercial GSM call marks 20-year milestone,” 2011 (1 June 1991 date; secondary). https://www.networkworld.com/article/2178889/first-commercial-gsm-call-marks-20-year-milestone.html

[13] Wikipedia, “GSM” – first GSM call 1 July 1991; 1987 first specification; GPRS 2000 (secondary).

[14] Wikipedia, “1G” – NTT 1979, NMT 1981, AMPS 1983; fragmentation; cellular concept (secondary). https://en.wikipedia.org/wiki/1G

[15] Wikipedia, “Mobile telephony” – MTS 1946, NMT roaming, Chicago 1983 (secondary). https://en.wikipedia.org/wiki/Mobile_telephony

[16] RCR Wireless, “Cellular under the tree in ’83: The first commercial cellular call in America” – MTS 17 June 1946; IMTS 1964. https://www.rcrwireless.com/?p=178647

[17] ITU-R, “IMT towards 2030 and beyond (IMT-2030)” portal – M.2160, six usage scenarios, submission timeline. https://www.itu.int/en/itu-r/study-groups/rsg5/rwp5d/imt-2030/pages/default.aspx

[18] ITU, “IMT-2030: Technical requirements for the 6G future,” March 2026 – 20 TPRs; approval expected in December. https://www.itu.int/hub/2026/03/imt-2030-technical-requirements-for-the-6g-future/

[19] IEEE ComSoc Technology Blog, “ITU-R: IMT-2030 (6G) Backgrounder and Envisioned Capabilities,” 2024 (secondary). https://techblog.comsoc.org/2024/07/06/itu-r-imt-2030-6g-backgrounder-and-envisioned-capabilities/

[20] “6G: The Intelligent Network of Everything,” arXiv:2407.09398 – peak/user-experienced rate definitions and IMT-2030 values (secondary). https://arxiv.org/pdf/2407.09398

[21] University of Pittsburgh course slides, “Intro to 3G Cellular Systems and UMTS Overview” – typical 2.5G rates 30–70 kbps (secondary). https://sites.pitt.edu/~dtipper/2720/2720_Slides11.pdf

[22] Microwaves & RF, “Bracing for the cellular explosion” – NTT 1979; all-IP 4G (secondary). https://mwrf.com/markets/article/21841593/bracing-for-the-cellular-explosion

[23] Wikipedia, “2G” – encryption (radio link only), spectrum efficiency, GPRS/EDGE rates (secondary).

[24] 3G4G / Intel LTE presentation – HSPA+ peak rates 14.4/28.8/43.2 Mbps (secondary). https://www.3g4g.co.uk/Lte/LTE_Pres_0805_Intel.pdf

[25] Wikipedia, “IMT-2020” – reference list entry “LTE-M and NB-IoT meet the 5G performance requirements” (ITU); article not opened (secondary).